GitHub Repository Guidelines

- Create one parent GitHub repository named:

  👉 **RollNo-AI-Your_Name**

  Inside the repository, create separate folders for each laboratory session, such as:

  - Lab1
  - Lab2
  - Lab3
  - Lab4
  - ...
  - LabN
- Each lab folder should contain:
  - Python source files (`.py`) or Jupyter/Colab notebooks (`.ipynb`)
  - Input datasets (if applicable)
  - Generated output files/visualizations
  - **A `README.md` including:**
    - Aim / Problem Statement
    - Algorithm
    - Results / Output
    - Performance Analysis (Accuracy / Precision / Recall / F1-score / Execution Time, wherever applicable)
    - Screenshots / Graphs / Confusion Matrix / Model Visualization
    - Learning Outcomes
- Follow good programming practices:
  - Write modular and reusable code.
  - Use meaningful variable, class, and function names.
  - Add appropriate comments and documentation.
  - Handle exceptions and invalid inputs.
  - Follow PEP-8 coding standards (for Python).
  - Organize code into functions/classes where appropriate.
- **Create a `datasets/` folder** (shared datasets used across experiments).
- **Create a `models/` folder** (trained models, serialized weights, checkpoints, if applicable).
- **Create a `docs/` folder containing:**
  - algorithm-flowcharts/
  - architecture-diagrams/
  - performance-analysis.pdf
  - mini-project-report.pdf

    - presentation.pptx (optional)

- Use Git effectively:
    - Create branches for major experiments or mini-projects.
    - Write meaningful commit messages.
    - Push changes regularly after completing each laboratory exercise.
    - Tag major milestones (e.g., Mid-Sem, Final Submission).

Suggested Lab-wise Mapping	
Lab	**Experiment**
Lab 1	AI Environment Setup, Production Systems, Water Jug Problem
Lab 2	BFS, DFS, Hill Climbing, Best First Search, A*
Lab 3	Data Preprocessing, EDA, NumPy, Pandas, Visualization
Lab 4	Knowledge Representation, Rule-Based Systems, Semantic Networks
Lab 5	Decision Tree Classifier, Model Evaluation
Lab 6	PyTorch Basics, Linear Regression, Neural Network Fundamentals
Lab 7	Explainable AI (Grad-CAM), Adversarial AI (FGSM), PCA/SVD, Mini Project

RollNo-AI-Your_Name/

```
├── Lab1/
│   ├── Experiment1.ipynb
│   ├── Experiment2.py
│   ├── outputs/
│   └── README.md
│
├── Lab2/
│   ├── SearchAlgorithms.ipynb
│   └── README.md
│
├── Lab3/
│   ├── DataPreprocessing.ipynb
│   └── README.md
│
├── Lab4/
│   ├── KnowledgeRepresentation.ipynb
│   └── README.md
│
├── Lab5/
│   ├── DecisionTree.ipynb
│   └── README.md
│
├── datasets/
│   ├── placement.csv
│   ├── iris.csv
│   └── ...
│
├── models/
│   ├── decision_tree.pkl
│   ├── linear_regression.pt
│   └── ...
│
├── docs/
│   ├── algorithm-flowcharts/
│   ├── architecture-diagrams/
│   ├── performance-analysis.pdf
│   ├── mini-project-report.pdf
│   └── presentation.pptx
│
├── README.md       (Repository Overview)
├── requirements.txt
├── LICENSE         (Optional)
└── .gitignore
```

[Analysis and Application part to be done by students as Homework Assignment]

Unit 1: AI Problem Formulation & Intelligent Problem Solving (CO1)

Basic	➔ AI Development Environment (Python, Jupyter, Google Colab) ➔ Problem Formulation in AI ➔ State Space Representation ➔ Production Systems ➔ Rule-Based Reasoning ➔ Problem Characteristics
Laboratory Activities	➔ Implement Production Rule Systems. ➔ Water Jug Problem. ➔ Missionaries and Cannibals Problem. ➔ Vacuum Cleaner Problem. ➔ State Space Representation. ➔ Rule-based Decision Making.
Application	★ Rule-Based Medical Diagnosis System ★ Intelligent Chatbot using Production Rules
Analysis	★ Compare Classical Programming and AI Programming. ★ Analyze different AI problem formulations.

Unit 2: Heuristic Search Algorithms (CO2)

Basic	➔ Generate and Test ➔ Breadth First Search (BFS) ➔ Depth First Search (DFS) ➔ Hill Climbing ➔ Best First Search ➔ A* Search
Laboratory Activities	➔ Maze Solver ➔ Graph Traversal ➔ Route Planning ➔ Robot Navigation ➔ 8-Puzzle Solver
Application	★ GPS Route Planner ★ Intelligent Maze Solver

Analysis	★ Compare heuristic search algorithms using execution time and solution quality.

Unit 3: AI Programming Frameworks & Data Preparation (CO2, CO5)

Basic	➔ NumPy for numerical computing ➔ Pandas for data preprocessing ➔ Matplotlib for visualization ➔ Scikit-learn overview ➔ Dataset loading and preprocessing ➔ Feature Engineering ➔ Data Encoding ➔ Train-Test Split
Laboratory Activities	➔ Dataset Loading ➔ Exploratory Data Analysis (EDA) ➔ Data Cleaning ➔ Missing Value Handling ➔ Label Encoding ➔ Feature Selection ➔ Data Visualization
Application	★ Student Placement Dataset Analysis ★ Customer Analytics ★ Data Preprocessing Pipeline
Analysis	★ Compare preprocessing techniques. ★ Evaluate impact of feature engineering.

Notebook References	ML_Libraries.ipynb
	Assignment-01a.ipynb
	MML_NumPy_Maths_Notebook.ipynb

Unit 4: Knowledge Representation & Logical Reasoning (CO1, CO3)

Basic	➔ Rule-Based Knowledge Representation ➔ Predicate Representation ➔ Semantic Networks

	➔ Frames ➔ Conceptual Dependency ➔ Logical Inference
Laboratory Activities	➔ University Knowledge Base ➔ Medical Knowledge Representation ➔ Semantic Network Construction ➔ Frame Representation ➔ Logical Rule Engine
Application	★ Expert System Knowledge Base ★ Intelligent Recommendation Engine
Analysis	★ Compare Frames, Semantic Networks and Rule-Based Systems.

Unit 5: Machine Learning for Intelligent Decision Making (CO4, CO5)

Basic	➔ Feature Matrix Creation ➔ Model Training ➔ Decision Tree Classifier ➔ Prediction ➔ Performance Evaluation ➔ Confusion Matrix ➔ Classification Report
Laboratory Activities	➔ Student Placement Prediction ➔ Disease Prediction ➔ Loan Approval Prediction ➔ Decision Tree Visualization
Application	★ Intelligent Decision Support System
Analysis	★ Evaluate Decision Tree performance. ★ Compare different evaluation metrics.

Unit 6: Deep Learning Fundamentals using PyTorch (CO4, CO5)

Basic	➔ Tensor Operations ➔ Automatic Differentiation ➔ Linear Regression using PyTorch ➔ Neural Network Basics ➔ Model Training

	➔ Loss Functions ➔ Optimizers
Laboratory Activities	➔ Tensor Manipulation ➔ Linear Regression ➔ Training Loop ➔ Gradient Computation ➔ Model Evaluation
Application	★ House Price Prediction ★ Student Score Prediction ★ Sales Forecasting
Analysis	★ Compare traditional ML and Deep Learning models. ★ Analyze convergence of gradient-based learning.

Notebook References	Pytorch_Basics.ipynb
	Pytorch_Basics(1).ipynb
	Linear_regression_pytorch.ipynb
	Linear_regression_pytorch(1).ipynb
	Linear_regression_pytorch_II.ipynb

Unit 7: Explainable AI, Adversarial AI & Advanced Applications (CO4, CO5)

Basic	**Explainable AI** ➔ Grad-CAM Visualization ➔ Feature Importance ➔ Model Interpretability **Laboratory Activities** ➔ CNN Visualization using Grad-CAM ➔ Adversarial Attack using FGSM ➔ ResNet-based Image Classification ➔ PCA for Dimensionality Reduction	**CNN Applications** ➔ Image Classification ➔ Transfer Learning ➔ Model Visualization **Adversarial AI** ➔ Fast Gradient Sign Method (FGSM) ➔ Adversarial Image Generation ➔ Robustness Evaluation

	➔ Feature Visualization	
Application	**Develop any one** ● Medical Image Classification ● Intelligent Traffic Sign Recognition ● Face Recognition Prototype ● Explainable AI Dashboard ● Adversarial Attack Detection System	
Analysis	● Compare Explainable AI techniques. ● Analyze robustness against adversarial attacks. ● Compare PCA with feature extraction techniques.	

Notebook References	GradCAM.ipynb
	FastGradientSignMethod-FGSM.ipynb
	resnet-ceee-lab3.ipynb
	MML_SVD_PCA.ipynb